

OneCaroyroyan: A Web-Based Barangay E-Service and Management Information System

Web Application Proposal

Area of Application: Transaction Processing System (TPS) and
Service Operations Management System

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BACKGROUND OF THE STUDY

Barangay Caroyroyan is a frontline local government unit responsible for providing essential administrative and community services to its residents. These services include the issuance of Barangay Clearance, Certificate of Residency, Certificate of Indigency, Cedula, Business Clearance, and the recording of blotter and complaint cases. These documents are vital for employment, education, legal transactions, and access to government assistance.

At present, Barangay Caroyroyan currently operates through a walk-in-only service process. Residents personally visit the barangay hall to request services, which are processed through a counter-based workflow. Certificate processing is handled efficiently, with an average processing time of less than ten (10) minutes per request. During peak days, queues sometimes occur, with senior citizens being the most affected.

Resident records are currently stored using physical folders and Excel spreadsheets, allowing staff to retrieve records very easily. While the barangay does not experience frequent record loss, there have been instances of damaged documents due to reliance on physical storage. Certificates are prepared using standardized templates, but the process is not yet automated and still requires manual encoding.

Financial transactions are recorded through logbooks, and spreadsheets. Although collection tracking is manageable, the Barangay Captain acknowledged that a more structured digital system could further improve organization and documentation. The barangay is not yet open to online payments, preferring to retain the Accounting Clerk's role to ensure accountability and compliance with financial procedures.

Blotter and complaint cases are recorded in logbooks and computer files, and their status is generally easy to monitor. However, a system with centralized storage, search, and filtering features was considered useful for improving case management.

Monthly reports are prepared using Excel computation, typically taking less than one hour, with minimal errors. Despite this, the Barangay Captain emphasized that an automated reporting system with dashboards and charts would be very helpful for monitoring operations and supporting decision-making.

The Barangay Captain highlighted that any proposed system should be orderly, systematic, procedural, and aligned with the mandates of the Department of the Interior and Local Government (DILG). In response, the researchers propose OneCaroyroyan: A Web-Based Barangay E-Service and Management Information System, designed to support existing walk-in transactions through internal digitalization, enhance record management, improve reporting, and strengthen governance without disrupting current operations.

PROBLEM STATEMENT

Barangay Caroyroyan currently delivers services through an efficient walk-in and counter-based process supported by manual and semi-digital record-keeping methods. While daily operations are manageable, several limitations remain due to the absence of a fully integrated management system.

Specifically, the barangay faces the following concerns:

- Dependence on physical records, which exposes documents to possible damage
- Manual preparation of certificates using templates without automation
- Fragmented record storage across folders and spreadsheets
- Limited centralized monitoring of transactions, collections, and complaints
- Manual report preparation that lacks visual analytics for decision-making

Although the current system functions adequately, the Barangay Captain recognizes the need for a more orderly, systematic, and procedural approach aligned with DILG mandates. Therefore, there

is a need to develop a Web-Based Barangay E-Service and Management Information System that enhances internal operations, improves record security, and supports data-driven governance.

OBJECTIVES OF THE STUDY

General Objective

The general objective of this study is to design and develop OneCaroyroyan: A Web-Based Barangay E-Service and Management Information System that enhances barangay operations by making them more orderly, systematic, and aligned with DILG procedural standards.

Specific Objectives

- To support walk-in transactions through a digital internal processing system
- To centralize resident records for better organization and protection
- To semi-automate certificate generation using standardized templates
- To improve financial documentation through structured digital recording
- To enhance monitoring of blotter and complaint cases through centralized records
- To generate automated reports and dashboards for barangay officials
- To support transparent and procedural barangay governance

SCOPE AND LIMITATIONS

Scope of the Study

This study focuses on the design and development of OneCaroyroyan, a web-based internal management system for Barangay Caroyroyan. The system is intended to improve record-keeping, document processing, and reporting for authorized barangay personnel while maintaining the current walk-in service workflow.

The system will:

- Provide a secure web-based platform accessible only to authorized barangay personnel.
- Support walk-in service requests
- Implement role-based access reflecting the current structure:
 - Accounting Clerk
 - Treasurer
 - Administrator
 - Barangay Secretary
 - Barangay Captain
- Manage resident records through a centralized database
- Semi-automate the generation of:
 - Barangay Clearance
 - Certificate of Residency
 - Certificate of Indigency
 - Cedula
 - Business Clearance

(Certificates are generated using templates but still issued manually at the counters.)
- Record and monitor financial transactions (cash-based) while keeping the roles of Accounting Clerk and Treasurer intact.
- Manage blotter and complaint records with status tracking (Pending, Ongoing, Settled).
- Generate automated reports and dashboards for resident transactions, certificate issuance, financial collections, and complaints.
- Allow report viewing and printing for documentation purposes.
- Implement a Hybrid Correspondence and Document Tracking module to log physical letters from walk-in external sectors and support direct digital uploads from LGU or National Agencies.

Limitations of the Study

While the proposed system aims to improve barangay operations, the study is subject to the following limitations:

- Online submission is limited only to official correspondence from external sectors (LGU/National Agencies) and will not be available for resident certificate requests.
- The system is intended solely for internal management at Barangay Caroyroyan and will not be available to other barangays.
- The system's effectiveness depends on proper staff training and stable internet connectivity.

TARGET CLIENTS AND USERS

Target Client

Barangay Caroyroyan (Local Government Unit)

System Users and Roles

- **Accounting Clerk**
 - Generate and issue the official receipts through the system upon recording of payment.
 - Works under the Treasurer's supervision.
 - Updates payment records for financial tracking.
- **Treasurer**
 - Validates and monitors financial transactions.
 - Generates monthly financial reports.
 - Oversees Accounting Clerk and encoder in the Treasurer's Office.
- **Administrator**
 - Processes certificate requests from residents.
 - Uses templates to semi-automate certificate generation.
 - Issues certificates to residents after verification.

- **Barangay Secretary**

- Receives and digitizes communications and letters from LGU, national, and external offices.
- Encodes and monitors complaints or blotter cases.
- Coordinates with Kagawad for complaint resolution.

- **Barangay Captain**

- Monitors summary reports and dashboards.
- Uses analytics for informed decision-making and governance.

PROPOSED SYSTEM FEATURES

1. Operational Features (Web System)

1.1 Role-Based User Authentication

The system shall provide secure login and logout functionality for all users. Access permissions shall vary based on assigned roles.

1.1.1 The system shall allow the Administrator to log in and access administrative and system configuration features, also process certificate requests.

1.1.2 The system shall allow the Barangay Secretary to log in and access resident records, encode/monitor complaints and blotter cases and manage official correspondence from external sectors.

1.1.3 The system shall allow the Accounting Clerk to log in and access payment recording modules, record cash transactions, and generate receipts under Treasurer supervision.

1.1.4 The system shall allow the Treasurer to log in and access payment verification, monitoring, and financial reporting modules.

1.1.5 The system shall allow the Barangay Captain to log in and access dashboards, summary reports, and analytics for decision-making.

1.2 User Account Management

The system shall allow management of user accounts with the following capabilities.

1.2.1 The Administrator shall add, edit, deactivate, or delete user accounts,

assign and modify user roles and approve new staff accounts. Additionally, the Administrator shall process and manage walk-in certificate requests.

1.2.2 The Barangay Secretary shall update their profile information and manage resident records, official correspondence, and complaint modules assigned to their roles.

1.2.3 The Accounting Clerk shall update their profile and manage cash payment modules.

1.2.4 The Treasurer shall update their profile and manage payment validation and financial reporting modules.

1.3 Resident Information Management

The system shall provide a centralized database for resident information and service transactions.

1.3.1 Authorized users shall add, edit, delete, and search resident records to ensure accurate and centralized data management.

1.3.2 The Administrator shall process requests for the following certificates, using templates to semi-automate document generation:

- Barangay Clearance
- Certificate of Residency
- Certificate of Indigency
- Cedula
- Business Clearance

1.3.3 The system shall record the issuance date, purpose, and the staff member responsible for each certificate.

1.4 Payment Recording System

The system shall allow the Accounting Clerk to record all cash payments, which will be verified by the Treasurer.

1.4.1 Payments shall include transaction details such as payer, certificate type, and amount.

1.4.2 Receipts shall be generated for residents for proof of payment.

1.4.3 Monthly financial reports shall be produced by the Treasurer based on recorded transactions.

1.5 Official Correspondence and Document Tracking

The system shall provide a centralized module for managing official letters and memorandums.

1.5.1 The system shall allow the Secretary to scan and log physical documents from walk-in external entities.

1.5.2 The system shall provide a Public Submission Portal for direct digital document uploads from LGU and National Agencies.

1.5.3 The system shall automatically generate an Electronic Tracking Number for every submitted document for status monitoring and accountability.

1.5.4 The system shall archive all correspondence in a dedicated data store (D4) for retrieval and review by the Barangay Captain.

1.6 Blotter and Complaint Management

The system shall allow authorized users to encode, track, and monitor complaints and blotter cases.

1.6.1 Users shall record complaint details including complainant, respondent, nature of complaint, and date filed.

1.6.2 Complaint status shall be updated as Pending, Ongoing, Settled, or Dismissed.

1.6.3 Authorized users shall be able to search, filter, and generate complaint reports for monitoring and management purposes.

2. MIS / Reporting Features (Management Information System)

2.1 Dashboard Overview

The system shall provide a real-time dashboard displaying key metrics for authorized users.

2.1.1 The Administrator shall view a comprehensive overview of resident records, certificate issuance, payments, and complaint status.

- 2.1.2 The Secretary shall view dashboards showing pending correspondence actions, resident record updates, and complaint monitoring relevant to their responsibilities.
- 2.1.3 The Accounting Clerk shall view dashboards displaying recorded cash payments for daily verification.
- 2.1.4 The Treasurer shall view dashboards showing daily, weekly, and monthly financial collections and verify transaction trends.
- 2.1.5 The Barangay Captain shall access dashboards summarizing issued certificates, financial collections, complaints, and real-time notifications of incoming official correspondence.

2.2 Automated Reporting

The system shall generate, view, and export reports for documentation and administrative use.

- 2.2.1 Reports shall summarize issued certificates monthly and yearly.
- 2.2.2 Reports shall summarize financial collections and cash transactions, including totals and breakdown per certificate type.
- 2.2.3 Reports shall summarize complaints and blotter cases, including complainant, respondent, nature of complaint, date filed, and status.
- 2.2.4 Reports shall be exportable in PDF format and printable for official use.

2.3 Charts and Analytics

The system shall provide visual analytics to support decision-making.

- 2.3.1 The Administrator shall view charts summarizing resident transactions, certificate requests, and payments.
- 2.3.2 The Secretary / Staff shall view graphs of certificate processing and complaint trends relevant to their role.
- 2.3.3 The Accounting Clerk and Treasurer shall view financial charts showing daily/monthly collections.
- 2.3.4 The Barangay Captain shall access charts summarizing issued certificates, collections, and complaints for strategic planning.

INITIAL SYSTEM OVERVIEW DIAGRAM

Context Diagram

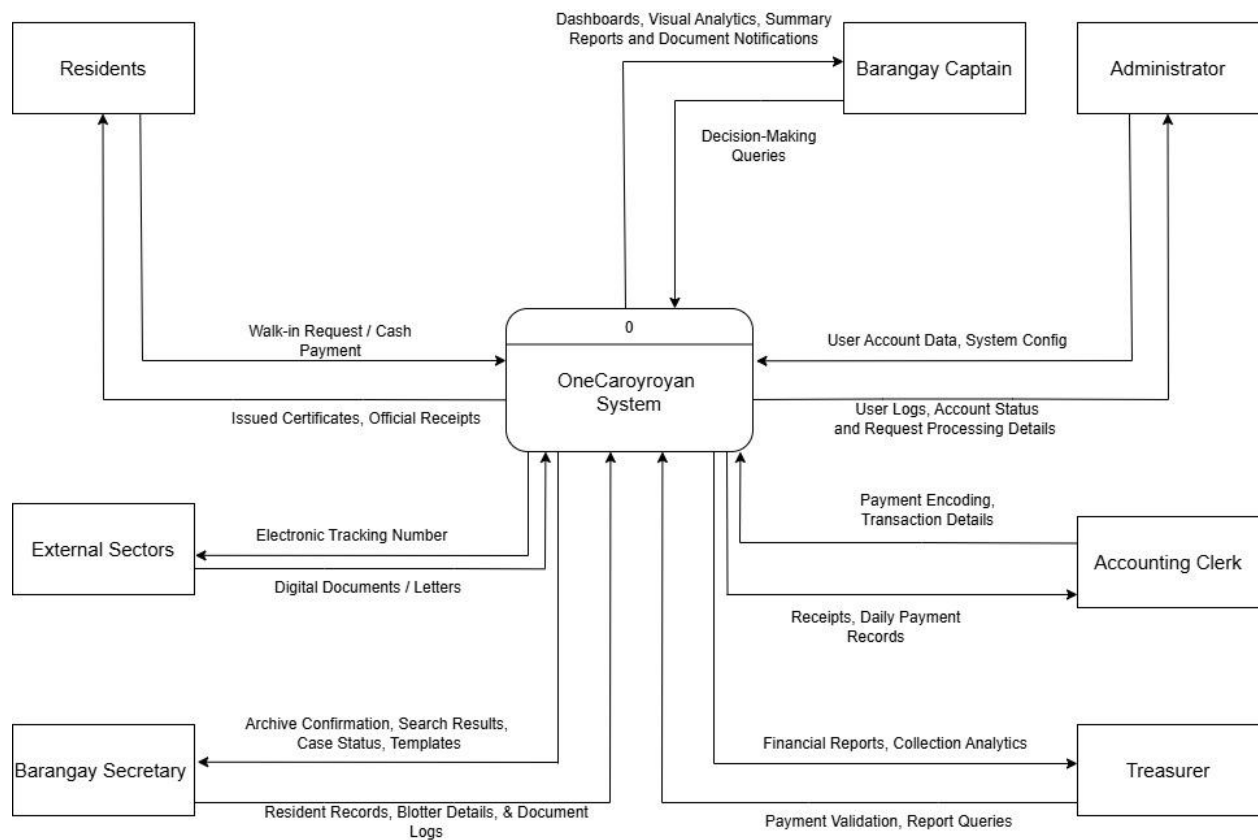


Figure 1: Context Diagram of the OneCaroyroyan System (Draw.io)

The Context Diagram (Figure 1) illustrates the high-level boundary of the OneCaroyroyan System and its interaction with six external entities, establishing the system as a centralized hub for barangay operations. It defines the flow of information between internal staff, residents, and external government sectors.

In this workflow, the system processes Walk-in Requests and Cash Payments from Residents, managed by the Administrator who inputs request details. The Barangay Secretary encodes resident records, blotter details, and document logs, while External Sectors provide digital

documents and receive electronic tracking numbers. The system then generates Issued Certificates and Official Receipts for residents, Financial Reports for the Treasurer, and real-time Dashboards and Notifications for the Barangay Captain.

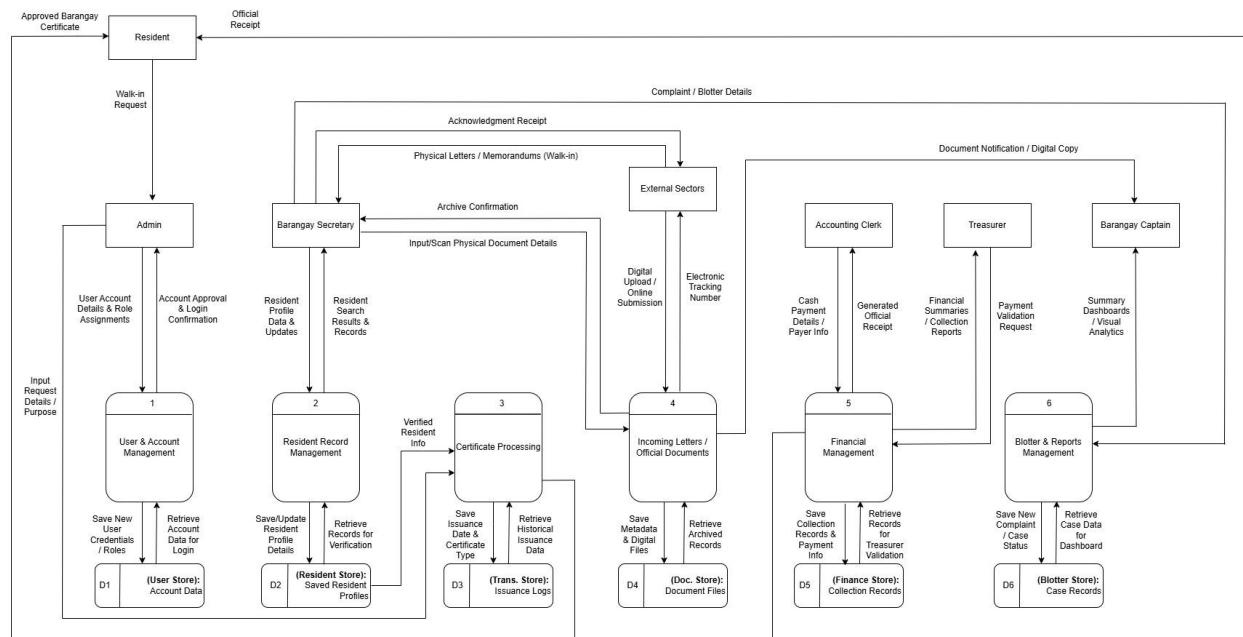


Figure 2 illustrates the decomposition of the OneCaroyroyan System into six core functional modules: User Management (1), Resident Management (2), Certificate Processing (3), Incoming Letters/Documents (4), Financial Management (5), and Blotter & Reports (6). This diagram defines the structured data exchange between external entities, internal processes, and the system's dedicated data stores (D1-D6).

CONCLUSION

After studying the current operations of Barangay Caroyroyan, we found that while their manual services are working, there is still a lot of room for improvement through technology. Our proposed system, OneCaroyroyan, was built not just to replace old logbooks, but to make every transaction in the barangay hall faster, safer, and more "procedural" for everyone involved.

Through this system, creating documents like Barangay Clearances and Certificates of Indigency will no longer require repetitive typing because of our Semi-Automated Templates. This will significantly reduce human errors in records and allow residents to get their needed documents much faster. In terms of Official Correspondence, letters and memorandums from the LGU and National Agencies will no longer be misplaced or forgotten because they will now have digital copies and tracking numbers that the Barangay Secretary can easily search in the database.

Another major benefit of OneCaroyroyan is the promotion of transparency and accountability. Since the Treasurer, Accounting Clerk, and Administrator each have their own secure accounts, it is very clear who received a payment and who issued a certificate. The Dashboards and Visual Charts will also serve as a guide for the Barangay Captain, allowing him to see the overall status of the barangay from financial collections to the number of blotter cases, without needing to spend hours on manual Excel computations.

In the end, OneCaroyroyan is our contribution to helping Barangay Caroyroyan become more ready for DILG mandates. We believe that when a barangay's internal management is orderly and systematic, the service they provide to every family in the community becomes better and more honest. This system is the first step toward a modern government that relies on accurate data and reliable technology to serve its people.